



UNIT I — INTRODUCTION TO CLOUD COMPUTING

Subject: Cloud Computing

Branch: Computer Engineering | Semester: V | SPPU 2024 Pattern

SYLLABUS TOPICS:

- Cloud Fundamentals: Definition, Importance of cloud computing
- Advantages and Disadvantages of Cloud Computing
- Characteristics
- Categories of Clouds: Private clouds, Public clouds
- Cloud Service Models: SaaS, PaaS, IaaS
- Cloud Architecture
- Cloud Storage: Distributed Data Storage, Data management
- Cloud Deployment Models

1.1 Cloud Computing : Definition and Importance

Definition: Cloud computing is a model that provides on-demand network access to a shared pool of configurable computing resources, such as servers, storage, networks and applications, which can be rapidly provisioned and released with minimal management effort.

Before cloud computing, every organisation had to buy its own servers, set up a data centre (a room full of computers that store and process data) and hire staff to maintain that hardware.

This approach needed heavy upfront investment and the capacity was often wasted because traffic on any application keeps changing throughout the year.

Cloud computing solves this problem by letting an organisation rent computing power, storage and software from a cloud provider over the internet, and pay only for what is actually used.

We study this topic because almost every modern IT system, from mobile apps to banking software, now runs partly or fully on cloud infrastructure, so it decides how systems are designed, deployed and secured.

1.1.1 Overview of Cloud Computing

A cloud computing system has two main parts working together over the internet: the front end (the part the user sees, such as a browser or mobile app) and the back end (the servers, storage and databases that actually do the work).

The back end is managed inside large data centres (buildings full of networked servers) that are owned by the cloud provider, and a special software layer called a hypervisor divides one physical server into many virtual machines so that many users can share the same hardware safely.

Requests from the client travel over the internet to the provider's data centre, get processed by the allocated virtual resources, and the result is sent back to the user, all within seconds.

Key sub-concepts of cloud computing overview:

- Cloud client - the device or software (browser, app, terminal) used to access cloud services.
- Distributed servers - many servers, often in different cities, that share the workload.
- Virtualization - the technique of creating multiple virtual machines on one physical server.
- Internet as the backbone - the network that connects the client to the remote data centre.

1.1.2 Importance of Cloud Computing

Cloud computing is important because it changes computing from a capital expense (buying hardware) into an operating expense (renting a service), which frees up money for the actual business.

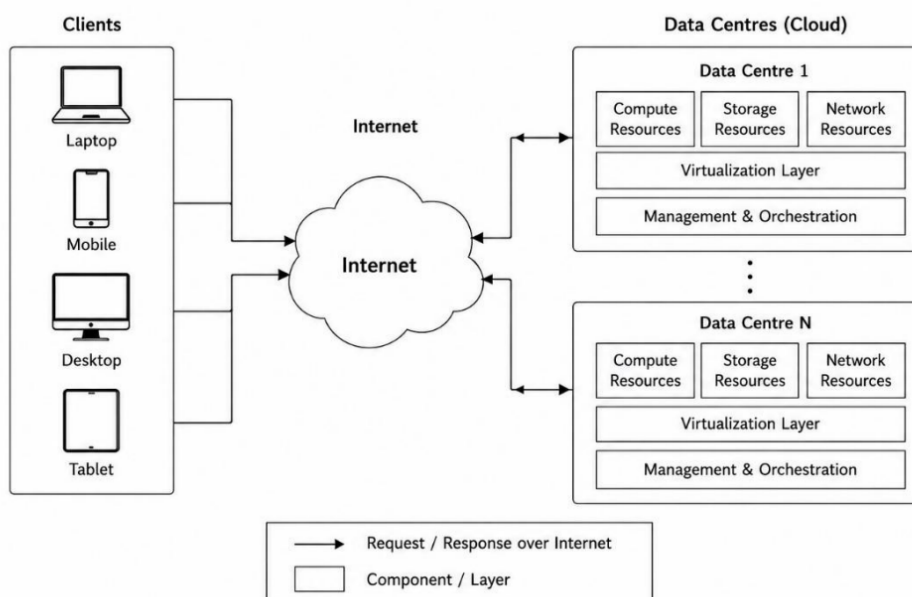
It also allows a small startup and a large enterprise to use the same world-class infrastructure, so competing on technology no longer depends on how much hardware a company can afford.

Why cloud computing matters today:

- Reduced capital expenditure - no need to buy physical servers or build a data centre.
- Faster time to market - a new application can be deployed in minutes instead of months.
- Global accessibility - services can be reached from anywhere with an internet connection.
- Built-in disaster recovery - provider maintains backups across multiple locations.
- Focus on core business - IT maintenance is handled by the provider, not the company.

FIGURE PLACEHOLDER

Fig 1.1 — Basic Cloud Computing Architecture showing Clients, Internet and Data Centres



Sr. No.	Parameter	Description
1	Traditional IT	Company buys and maintains its own servers and software licences.
2	Cloud Computing	Company rents computing resources from a provider over the internet.
3	Cost Model	Traditional IT is capital expense; cloud computing is pay-per-use operating expense.
4	Scaling Time	Traditional IT takes weeks to add hardware; cloud scales in minutes.

WORKED EXAMPLE 1

Given: A college wants to host its examination portal. Buying servers costs Rs. 6,00,000 with 5 year life, while a cloud plan costs Rs. 8,000 per month.

Find: Which option is cheaper for 3 years of use, and why is cloud still preferred despite the numbers?

Solution:

Step 1: Calculate traditional cost for 3 years = Rs. 6,00,000 (fixed, hardware already bought).

Step 2: Calculate cloud cost for 3 years = 8000×36 months = Rs. 2,88,000.

Step 3: Compare: Cloud (Rs. 2,88,000) is lower than traditional (Rs. 6,00,000) for this duration.

Step 4: Consider that cloud also removes maintenance staff cost and gives automatic scaling during exam load spikes.

Ans: Cloud computing is cheaper and more flexible for the college in this 3 year scenario.

Advantages:

- Lower upfront cost since hardware purchase is avoided.
- Elastic scaling handles sudden increase in users automatically.
- Provider handles software updates and security patches.
- Services can be accessed from any location with internet.

Disadvantages:

- Complete dependency on a stable internet connection.
- Recurring subscription cost can exceed hardware cost in the long run.
- Organisation has less direct control over the physical hardware.

Applications:

- Netflix uses cloud computing to stream video to millions of users worldwide.
- Zoom runs its video conferencing backend on cloud data centres.
- Small startups use Amazon Web Services instead of buying servers.
- Government e-governance portals like DigiLocker run on cloud infrastructure.

KEY POINTS — Cloud Computing Fundamentals

- Cloud computing turns hardware ownership into a rented, on-demand service.
- Remember the two building blocks: front end (client) and back end (provider data centre).
- Exam trap: students often confuse cloud computing with just 'internet storage' - it also includes processing power and software.
- Most confused pair: capital expenditure (buying) versus operating expenditure (renting) - cloud always favours the second.
- SPPU frequently asks to define cloud computing and list its importance in 5 or 6 marks questions.

SPPU EXPECTED QUESTIONS — Cloud Computing Fundamentals

- Q1. Define cloud computing and explain its basic working with a diagram. (6 marks)
- Q2. Explain the importance of cloud computing for modern organisations. (5 marks)
- Q3. Differentiate between traditional IT infrastructure and cloud computing. (6 marks)
- Q4. Describe the front end and back end components of a cloud system. (5 marks)
- Q5. Justify with a numerical example why cloud computing reduces cost for a small organisation. (8 marks)

1.2 Advantages and Disadvantages of Cloud Computing

Definition: The advantages and disadvantages of cloud computing describe the benefits an organisation gains and the risks or limitations it accepts when it moves its computing resources to a cloud provider.

Every technology decision involves trade-offs, and cloud computing is no different, so an engineer must know both sides before recommending it for a project.

This topic is studied on its own because SPPU examiners frequently ask students to justify when cloud computing should or should not be used, based on these points.

1.2.1 Advantages of Cloud Computing

The biggest advantage is cost saving, because the organisation pays only for the resources it actually consumes instead of buying fixed hardware.

Beyond cost, cloud computing gives flexibility, reliability and speed that are very difficult to achieve with in-house infrastructure alone.

Sr. No.	Advantage	Explanation
1	Cost Reduction	No upfront hardware purchase; pay-as-you-go pricing lowers total spending.
2	Scalability	Resources can be increased or decreased instantly to match demand.
3	Accessibility	Data and applications can be accessed from any device, anywhere.
4	Reliability	Provider maintains redundant servers, so downtime is minimised.
5	Automatic Updates	Software and security patches are applied by the provider automatically.

Sr. No.	Advantage	Explanation
6	Disaster Recovery	Data is backed up across multiple locations, protecting against loss.

1.2.2 Disadvantages of Cloud Computing

The main disadvantage is that cloud computing needs a working internet connection at all times, since the actual processing happens far away in the provider's data centre.

Organisations also give up some control over their own data and become dependent on a single vendor's policies and pricing.

Sr. No.	Disadvantage	Explanation
1	Internet Dependency	Without internet access, cloud based applications cannot be used.
2	Security and Privacy Risk	Sensitive data is stored on a third party's server, raising trust concerns.
3	Vendor Lock-in	Moving from one cloud provider to another can be costly and complex.
4	Limited Control	Customer cannot control the physical hardware or its exact location.
5	Downtime Risk	Outage at the provider's end affects every customer using that service.

Applications where this trade-off matters most:

- Startups adopt SaaS tools like Google Workspace to avoid IT overhead from day one.
- Banks weigh disadvantages like data privacy before moving core banking to public cloud.
- E-commerce platforms like Flipkart rely on cloud scalability during sale events.

KEY POINTS — Advantages and Disadvantages of Cloud Computing

- Cost saving and scalability are the two advantages examiners expect first.
- Security and internet dependency are the two disadvantages examiners expect first.
- Exam trap: do not only list advantages - questions often ask 'with justification', so always add one line of reasoning.
- Most confused pair: vendor lock-in (contractual/technical difficulty to switch) versus limited control (cannot manage own hardware).
- SPPU commonly frames this as a 5 mark 'list and explain' question.

SPPU EXPECTED QUESTIONS — Advantages and Disadvantages of Cloud Computing

- Q1. List and explain any five advantages of cloud computing. (5 marks)
- Q2. Explain the major disadvantages or limitations of cloud computing. (5 marks)
- Q3. Compare the advantages and disadvantages of cloud computing with suitable examples. (6 marks)
- Q4. Discuss the security and vendor lock-in risks associated with cloud adoption. (6 marks)

1.3 Characteristics of Cloud Computing

Definition: Characteristics of cloud computing are the essential properties, defined by NIST (National Institute of Standards and Technology), that a computing system must show before it can be called a true cloud service.

A system is not automatically a 'cloud' just because it uses the internet; it must show self-service, elasticity, pooling of resources, wide access and measured usage together.

SPPU asks this topic almost every year because it is the exact boundary that separates cloud computing from ordinary client-server or hosting services.

1.3.1 On-Demand Self-Service

A user can provision computing resources such as server time or storage automatically, without needing a human employee from the cloud provider to manually set it up.

This is normally done through a web console or an API (Application Programming Interface, a set of rules that lets software request services automatically), which reduces waiting time from days to minutes.

1.3.2 Broad Network Access

Cloud services must be available over the network and reachable using standard mechanisms, so that different client types like laptops, tablets and mobile phones can all connect.

This characteristic is what allows an employee to check a company dashboard equally well from a office desktop or a phone while travelling.

1.3.3 Resource Pooling

The provider's computing resources are pooled together to serve many customers using a multi-tenant model (many tenants sharing the same physical hardware safely), with resources assigned and reassigned dynamically.

The customer generally has no knowledge of the exact physical location of the resource, only that it is available when required, which is called location independence.

1.3.4 Rapid Elasticity

Capabilities can be elastically provisioned and released, in some cases automatically, so the system can quickly scale outward or inward as the workload changes.

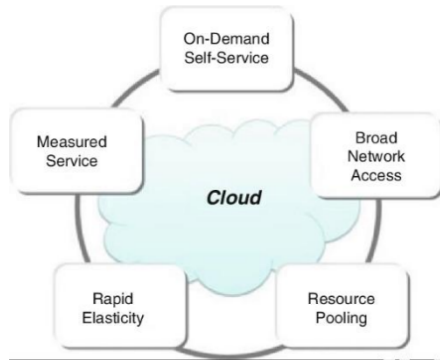
To the customer, the resources available for use often appear unlimited and can be requested in any quantity at any time, which is very useful for handling sudden traffic spikes.

1.3.5 Measured Service (Pay-per-use)

Cloud systems automatically control and optimise resource use by metering (measuring) usage, such as storage, processing, bandwidth and active user accounts.

Usage is monitored, controlled and reported, giving transparency for both the provider and the customer, and this metering is what makes pay-as-you-go billing possible.

Fig 1.2 — Five Essential Characteristics of Cloud Computing



Sr. No.	Characteristic	Benefit it Provides	Real Example
1	On-Demand Self-Service	No manual approval delay	Launching an AWS EC2 instance from console
2	Broad Network Access	Access from any device	Gmail available on mobile and desktop
3	Resource Pooling	Efficient hardware usage	Multiple tenants on one physical server
4	Rapid Elasticity	Handles sudden load spikes	Netflix scaling servers during peak hours
5	Measured Service	Fair, transparent billing	AWS billing based on exact CPU-hours used

WORKED EXAMPLE 2

Given: An e-commerce site normally uses 10 virtual servers, but during a festival sale the traffic increases 6 times for 2 days only.

Find: Which cloud characteristic solves this situation and how many servers are approximately needed at peak?

Solution:

Step 1: Identify the characteristic needed: rapid elasticity, since demand is temporary and sudden.

Step 2: Calculate approximate peak requirement = 10 servers x 6 = 60 servers.

Step 3: Cloud provider auto-scales from 10 to 60 servers during the sale.

Step 4: After 2 days, elasticity releases extra servers back, so billing returns to normal.

Ans: Rapid elasticity allows scaling up to about 60 servers and back down without manual purchase.

Advantages:

- On-demand self-service removes dependency on provider's support staff.
- Elasticity avoids both under-provisioning and wasteful over-provisioning.
- Measured service gives accurate, transparent, itemised billing.
- Resource pooling improves hardware utilisation across many customers.

Disadvantages:

- Achieving true elasticity requires complex automation and monitoring systems.
- Multi-tenant resource pooling raises concerns about data isolation between customers.
- Measured billing can become unpredictable if usage is not monitored carefully.

Applications:

- Amazon Web Services Auto Scaling groups demonstrate rapid elasticity in production.
- Salesforce demonstrates resource pooling through its multi-tenant SaaS platform.
- Google Drive demonstrates broad network access across phones, tablets and laptops.
- AWS Billing Console demonstrates measured service with per-second usage tracking.

KEY POINTS — Characteristics of Cloud Computing

- Remember all five NIST characteristics using the short form: Self-service, Broad access, Pooling, Elasticity, Measured service.
- Exam trap: students often forget 'measured service' and only mention four characteristics.
- Most confused pair: resource pooling (sharing hardware) versus rapid elasticity (scaling capacity up/down).
- SPPU frequently asks to 'list and explain characteristics with example' for 6 marks.
- Multi-tenancy is the underlying mechanism that makes resource pooling possible.

SPPU EXPECTED QUESTIONS — Characteristics of Cloud Computing

- Q1. List and explain the essential characteristics of cloud computing as per NIST. (6 marks)
- Q2. Explain rapid elasticity with a suitable real-world example. (5 marks)
- Q3. Describe on-demand self-service and measured service in cloud computing. (5 marks)
- Q4. Differentiate between resource pooling and rapid elasticity. (6 marks)
- Q5. Explain how essential characteristics of cloud computing support scalable applications. (8 marks)

1.4 Categories of Clouds : Private Clouds, Public Clouds

Definition: Categories of clouds classify cloud infrastructure based on ownership and accessibility: a private cloud is dedicated to a single organisation, while a public cloud is shared among multiple customers over the open internet.

Choosing between private and public cloud is one of the first architectural decisions any organisation must make, since it affects cost, security and control together.

This classification also forms the base for the complete set of deployment models (hybrid and community) that are studied later in Section 1.8 of this unit.

1.4.1 Private Cloud

A private cloud is cloud infrastructure provisioned for exclusive use by a single organisation, which may be owned, managed and operated by the organisation itself, a third party, or a combination of both.

Because only one organisation uses the hardware, a private cloud gives more control and can be customised to meet strict compliance or security requirements, such as those needed in banking or defence.

Sub-concepts of private cloud:

- On-premise private cloud - hosted inside the organisation's own data centre.
- Hosted private cloud - hardware is dedicated but physically located at a third-party provider.
- Higher cost per user since infrastructure is not shared with other customers.
- Best suited where data sensitivity or regulation demands full control.

1.4.2 Public Cloud

A public cloud is cloud infrastructure provisioned for open use by the general public, and it is owned, managed and operated by a business, academic or government organisation, or some combination of them.

The infrastructure physically exists on the premises of the cloud provider, and the same hardware is shared securely among many different customers using multi-tenancy.

Sub-concepts of public cloud:

- Multi-tenant architecture shares hardware efficiently among thousands of customers.
- Lower entry cost since customers only pay for what they consume.
- Provider handles all maintenance, security patching and hardware failures.
- Best suited for startups, websites and applications with variable or unknown load.

Sr. No.	Parameter	Private Cloud	Public Cloud
1	Ownership	Single organisation	Cloud service provider
2	Cost	High initial investment	Low, pay-per-use
3	Security	Higher, dedicated resources	Shared, provider managed
4	Scalability	Limited by owned hardware	Very high, provider scale
5	Example	Bank's internal cloud	Amazon Web Services, Microsoft Azure

WORKED EXAMPLE 3

Given: A hospital needs to store confidential patient records and also wants to run a public appointment booking website.

Find: Which type of cloud should be used for each requirement?

Solution:

WORKED EXAMPLE 3

Step 1: Analyse the patient records requirement: high sensitivity, needs strict compliance.

Step 2: Assign private cloud for patient records due to full control over security.

Step 3: Analyse the appointment booking website: variable public traffic, low sensitivity.

Step 4: Assign public cloud for the website since it benefits from elasticity and low cost.

Ans: Patient records go on a private cloud; the booking website is hosted on a public cloud.

Advantages:

- Private cloud offers maximum control and customisation for sensitive workloads.
- Public cloud offers very low entry cost with instant scalability.
- Both categories remove the need to predict exact future hardware needs.
- Both allow faster deployment compared to fully manual, on-premise setups.

Disadvantages:

- Private cloud still needs significant upfront capital for hardware.
- Public cloud gives less control over the exact physical infrastructure.
- Security responsibilities can become unclear when mixing categories without planning.

Applications:

- Banks and defence organisations use private clouds for core transaction systems.
- Government Aadhaar-linked systems often use private or hosted private clouds.
- Gmail and Netflix run entirely on public cloud infrastructure.
- Startups use AWS or Azure public cloud to avoid any hardware purchase.

KEY POINTS — Categories of Clouds

- Private cloud = single tenant, full control, high cost; Public cloud = multi-tenant, shared, low cost.
- Exam trap: private cloud does not always mean 'on company premises' - it can be hosted by a third party too.
- Most confused pair: private cloud versus hybrid cloud (hybrid combines both, covered in Section 1.8).
- Multi-tenancy is the key technical concept that enables public cloud's low pricing.
- SPPU frequently asks a direct comparison table question on this topic.

SPPU EXPECTED QUESTIONS — Categories of Clouds

- Q1. Define private cloud and public cloud with suitable examples. (5 marks)
- Q2. Differentiate between private cloud and public cloud. (6 marks)
- Q3. Explain the architecture of a private cloud with a diagram. (6 marks)

SPPU EXPECTED QUESTIONS — Categories of Clouds

Q4. Discuss situations where a private cloud is preferred over a public cloud. (5 marks)

Q5. With justification, decide which cloud category suits a given organisational scenario. (8 marks)

1.5 Cloud Service Models : SaaS, PaaS, IaaS

Definition: Cloud service models describe the layer of computing that a provider delivers to the customer as a service - infrastructure, platform, or complete software - and correspondingly, how much of the technology stack the customer must manage.

Every cloud offering can be placed at one of three layers of a technology stack: raw hardware, a development platform, or a finished application, and each layer transfers a different amount of management responsibility from the customer to the provider.

This topic is one of the most important in the entire subject because almost every SPPU question paper on cloud computing contains at least one direct question on service models.

1.5.1 Infrastructure as a Service (IaaS)

IaaS provides fundamental computing resources such as virtual machines, storage, and networks, where the customer does not manage the underlying physical hardware but has control over operating systems, storage and deployed applications.

The provider is responsible only for the physical servers, storage arrays, networking equipment and the virtualization layer, while the customer installs and manages everything above that, including the operating system.

Sub-concepts of IaaS:

- Virtual machines - customer chooses CPU, RAM and disk size as needed.
- Elastic storage - block or object storage that grows with the customer's data.
- Virtual networking - customer configures firewalls, subnets and IP ranges.
- Full OS-level control - customer installs any operating system or software stack.
- Billed typically by compute-hour, storage-GB, and data transfer.

1.5.2 Platform as a Service (PaaS)

PaaS provides a ready-made platform, including operating system, runtime environment, database and development tools, so the customer can deploy applications without managing servers directly.

The provider manages everything from hardware up to the runtime environment, while the customer focuses only on writing code and managing application data, which greatly speeds up development.

Sub-concepts of PaaS:

- Runtime environment - pre-configured language support such as Java, Python or Node.js.
- Managed database services - customer gets a ready database without server setup.
- Automatic scaling of the application tier as traffic changes.

- Built-in development and deployment tools reduce time to market.
- Customer has no access to the underlying operating system.

1.5.3 Software as a Service (SaaS)

SaaS delivers a complete, ready-to-use software application over the internet, where the customer only uses the application through a browser or client app and never manages any part of the underlying stack.

The provider is fully responsible for hardware, platform, application code, security patches and upgrades, and the customer typically pays a subscription fee per user or per feature tier.

Sub-concepts of SaaS:

- Multi-tenant application - one codebase serves all customers simultaneously.
- Subscription based billing - monthly or yearly fee per user seat.
- Zero installation - accessed directly through a web browser.
- Automatic feature updates roll out to all users without manual action.
- Examples include email, CRM, office productivity and video conferencing tools.

1.5.4 Comparison of IaaS, PaaS and SaaS

Although all three models are delivered 'as a service', they differ sharply in what the customer manages versus what the provider manages, and this difference decides which model suits which project.

A simple way to remember this is that IaaS gives the most control and the most responsibility, while SaaS gives the least control but also the least responsibility.

Sr. No.	Parameter	IaaS	PaaS	SaaS
1	What is provided	Virtual machines, storage, network	Runtime, database, dev tools	Complete application
2	Managed by customer	OS, middleware, application	Application and data only	Nothing, only usage
3	Managed by provider	Hardware, virtualization	Hardware up to runtime	Entire stack
4	Typical user	System administrators	Application developers	End users
5	Example	Amazon EC2, Google Compute Engine	Google App Engine, Windows Azure	Gmail, Salesforce, Dropbox
6	Billing basis	Per hour of compute/storage used	Per app instance or request	Per user per month

- Google App Engine is a PaaS used to deploy scalable web applications quickly.
- Microsoft Azure App Service is a PaaS used for enterprise web application hosting.
- Salesforce is a SaaS platform used worldwide for customer relationship management.
- Dropbox is a SaaS product used for file storage and sharing without any server setup.

KEY POINTS — Cloud Service Models

- Remember the responsibility gradient: IaaS (most customer control) to SaaS (least customer control).
- Exam trap: students often swap PaaS and IaaS examples - Google App Engine is PaaS, not IaaS.
- Most confused pair: PaaS versus SaaS, since both feel 'managed' - PaaS is for developers to build apps, SaaS is the finished app.
- The shared responsibility model is a favourite diagram-based question in SPPU papers.
- Always connect each model to one real example while answering - it improves marks significantly.

SPPU EXPECTED QUESTIONS — Cloud Service Models

- Q1. Explain IaaS, PaaS and SaaS with suitable examples. (6 marks)
- Q2. Differentiate between IaaS, PaaS and SaaS. (8 marks)
- Q3. Describe the shared responsibility model across the three cloud service models with a diagram. (8 marks)
- Q4. Explain Platform as a Service with its features and one real example. (6 marks)
- Q5. For a given scenario, select and justify the most suitable cloud service model. (8 marks)

1.6 Cloud Architecture

Definition: Cloud architecture is the overall design of a cloud computing system, describing how its front end, back end, network and internal layers are organised and connected to deliver services reliably to many users at once.

A well designed architecture decides how well a cloud system can handle failures, sudden traffic growth and security threats, so it is the blueprint every cloud engineer must understand before building anything on the cloud.

SPPU treats this as a core design topic, frequently asking students to draw and label the logical architecture or the reference model from memory.

1.6.1 Cloud Computing Logical Architecture

The logical architecture of a cloud system is divided into a front end and a back end, connected through the internet, where the front end is the part visible to the user and the back end is the actual cloud itself.

The front end includes the client's device and the applications needed to access the cloud system, such as a web browser, while the back end includes the physical servers, storage devices, virtual machines, and the security mechanisms that protect them.

Sub-concepts of logical architecture:

- Client interface - the browser, mobile app or thin client used to send requests.
- Middleware - manages communication and coordination between client and servers.
- Central server - allocates and monitors resources across the whole system.
- Security layer - authentication, authorization and encryption across the path.
- Management software - governs load distribution among many back-end servers.

1.6.2 Cloud Computing Reference Model

The reference model organises a cloud system into stacked layers, typically infrastructure, platform and application, sitting above a virtualization layer that abstracts the physical servers, storage and datacentres.

Each higher layer builds upon the services offered by the layer below it, so the infrastructure layer offers virtual machines, the platform layer offers a runtime, and the application layer offers the finished software to end users.

Sub-concepts of the reference model:

- Virtualization layer - creates virtual servers/storage from physical hardware.
- Infrastructure layer - exposes virtual machines, storage and network as services.
- Platform layer - exposes a runtime and managed services for developers.
- Application layer - exposes the finished software directly to end users.
- Cross-cutting management - monitoring, billing and security span all layers.

1.6.3 Cloud System Architecture

At an implementation level, cloud systems are commonly deployed using a single cloud site architecture, a redundant 3-tier architecture, or a multi-data centre architecture, depending on the scale and reliability needed.

A single cloud site architecture uses one DNS (Domain Name System, which converts a website name into a server address), one load balancer and one application-database pair, which is simple but has a single point of failure.

A redundant 3-tier architecture duplicates the load balancer, application and database tiers, using master-slave database replication, so that if one component fails the system continues to run using its backup.

A multi-data centre architecture goes further and spreads redundant 3-tier systems across two or more physical data centres in different locations, protecting the service even if an entire data centre goes offline.

Sub-concepts of cloud system architecture:

- DNS routes user requests to the nearest or healthiest available server.
- Load balancer distributes incoming traffic across multiple application servers.
- Master-slave database replication keeps a live backup copy of all data.
- Snapshot backups periodically save the entire state for disaster recovery.
- Geographic distribution across data centres reduces latency and increases uptime.